Greig Oldford

Science Manager · Research Lead · Science Advisor

BSc · Adv. Dip. GIS · MES · PhD candidate (UBC)

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PRE-QUALIFIED POOLS

PC-04	`21-DFO-NCR-EA-EOS-308850`	National Capital Region
PC-03	`18-DFO-PAC-IA-EFM-225908`	Pacific Region
BI-03	`2016 Pacific Region process`	Pacific Region

Current: Acting Section Head (BI-04), Nearshore Ecosystems, DFO Science, Pacific Region

Completed Section Head training and certifications

In process: DFO All Hands on Science 2.0, leadership stream (BI-04 / PC-04, internal) and BI-02/03 / PC-02/03 (public)

PROFILE

Versatile science leader with 13+ years at Fisheries and Oceans Canada in research, science management and advice, across both the Science and Ecosystem Management branches. I currently manage a section of research scientists and field and laboratory biologists. I have designed and led a funded ecosystem modelling program, led a 40-year Salish Sea ocean hindcast to publication, coordinated intergovernmental technical work for the Great Bear Sea MPA network with First Nations and the Province of BC, and given expert testimony to Parliament. I move easily between research, field programs, planning, policy and people management, and turn complex science into clear options for decision-makers.

AREAS OF STRENGTH

Research leadership	Designing and leading research programs, competitive funding, lead-authored papers, ocean and ecosystem model development, field and at-sea research
Science management	Budgets, staffing, performance management, field and at-sea programs, research planning aligned with departmental priorities
Science advice	Briefings for senior management, CSAS peer review (steering committee, subject-matter expert), parliamentary testimony
Partnerships	First Nations, Province of BC, ECCC, Parks Canada, universities, industry and stakeholders
Technical depth	Ecosystem and ocean modelling, GIS and spatial analysis, Python, R, HPC, marine spatial and conservation planning

SELECTED RESULTS

Recommended the Aquaculture Monitoring Program trial low-cost eDNA sampling of benthic communities under finfish farms; the advice secured budget and a pilot field season (July–September 2026).

Kept the Nearshore Ecosystems Section on track during budget cuts in 2026, completing the research cruise and field season, hiring co-op students and developing a new research program.

Won Competitive Science Research Fund support to start the Integrated Ecosystem Modelling program; hired and supervised a scientist and delivered the final report in 2025.

Led HOTSSea, a 40-year Salish Sea ocean hindcast, to publication (Geoscientific Model Development, 2025) after earlier attempts had stalled; it underpins the Strait of Georgia ecosystem model.

Co-led federal technical analyses, as co-chair of an intergovernmental technical working group, that informed the Great Bear Sea MPA network (endorsed 2023); team received the DFO Prix d'Excellence.

Expert witness to the House of Commons Fisheries Committee (2022); testimony fed into its 2023 report "Science at the Department of Fisheries and Oceans".

Hosted briefings and site tours of the Pacific Science Enterprise Centre for the Minister, Deputy Minister, ADM and Director General.

EXPERIENCE · FISHERIES AND OCEANS CANADA

2025–present

Science Manager / Section Head (A/BI-04)
Nearshore Ecosystems Section, Aquatic Diagnostics and Genomics Technologies, DFO Science, Pacific Science Enterprise Centre, West Vancouver

- Manage the programs of four research scientists and three field and laboratory biologists and technicians
- Responsible for section finances, hiring, performance management and goal-setting, collaborative agreements and field coordination; alternate senior manager for PSEC
- Scoped field program / research goals in line with a changing departmental mandate; briefed senior management

2024–2025

Science Program Lead, Integrated Ecosystem Modelling (BI-03)
Regional Ecosystem Effects on Fish & Fisheries Section, Ecosystem Science Division, DFO Pacific

- Designed a new research program, secured CSRF funding, and hired and supervised a scientist; acted as Section Head when required
- Led collaborations with other DFO groups, ECCC and universities
- Delivered management briefings and peer-reviewed papers; subject-matter expert in CSAS processes

2023–2024

Ecosystem Scientist and Modeller (BI-03)
Regional Ecosystem Effects on Fish & Fisheries Section, Ecosystem Science Division, DFO Pacific

- Lead developer of the Salish Sea ocean model (NEMO-SS1500) on Linux HPC; advice on early marine survival of coho and Chinook salmon

2022–2023

Senior Marine Spatial Planning Officer and Science Advisor (PC-03)
Integrated Oceans Management, Ecosystem Management Branch, DFO Pacific

- Represented DFO with First Nations, the Province and external groups; strategic input on marine conservation and climate science priorities

2021–2022

Researcher and Science Advisor (contract, education leave)
DFO Pacific and Parks Canada

- Led a synthesis of cross-sectoral data use in fisheries and oceans management; findings were used to brief DFO management
- Updated the FIPEX river-connectivity tool and analysed Ontario rivers for Parks Canada

2017–2018

Marine Spatial Planning Advisor (PC-03)
Integrated Oceans Management, Ecosystem Management Branch, DFO Pacific

- Managed technical aspects of MPA network planning for the Northern Shelf Bioregion; supervised casual staff
- Liaison between Science and Ecosystem Management; steering committee member for CSAS peer reviews; wrote and delivered briefings for senior management

2014–2017

Oceans Biologist (BI-02)
Integrated Oceans Management, Ecosystem Management Branch, DFO Pacific

- Led collaboration with provincial and Indigenous partners on MPA network planning
- Scoped CSAS requests for science advice; member of the Steering Committee; represented the DFO client for science advice
- Synthesised sensitive issues regarding science governance and delivered briefings to upper management

2013–2014

GIS Specialist and Programmer (BI-02)
Oceans & Coastal Management Division, DFO Maritimes

EDUCATION

expected 2026	**PhD, Oceans and Fisheries (ecosystem modelling)**, University of British Columbia
2013	**Master of Environmental Studies**, Dalhousie University
2007	**Advanced Diploma, Geographic Information Systems**, Nova Scotia Community College (COGS)
2004	**BSc, Environmental Science (Ecology)**, University of Guelph

AWARDS & FUNDING

2024	BC Premier's Award (team), Great Bear Sea MPA Network
2023	DFO Prix d'Excellence (team), Great Bear Sea MPA Network
2024–2025	DFO Competitive Science Research Fund (CSRF), integrated ecosystem modelling
2020	UBC President's Academic Excellence Initiative PhD Award
2019–2021	UBC Ocean Leaders Award & Fellowship

SCIENCE ADVICE & PUBLIC ENGAGEMENT

Jun 2026	What can ocean-ecosystem simulations tell us about long-term environmental change in the Strait of Georgia? *Sunshine Coast Ocean Festival (public talk)*, Sechelt, BC.
May 2026	Science, power, and decision-making. *Reimagining the Future of Oceans and Fisheries: UBC Institute for the Oceans and Fisheries 10th Anniversary Symposium (moderator: Nathan Bennett, WWF)*, Vancouver, BC.
Jan 2025	Insights from 'end-to-end' ecosystem modelling of the Strait of Georgia. *Salish Sea Science Roundtable, Puget Sound Institute (University of Washington), co-hosted with Fisheries and Oceans Canada*, Online.
May 2022	Expert witness, study of "Science at the Department of Fisheries and Oceans". *House of Commons Standing Committee on Fisheries and Oceans (FOPO), 44th Parliament, Meeting 21*, Ottawa, ON.

SELECTED PUBLICATIONS

Beaty, F., Martone, R. G., Cristiani, J., **Oldford, G.**, Bodtker, K., Robb, C., Rubidge, E., Gale, K., Yang, K., Brown, K. H. T., Schubert, A., & McDougall, C. (2026, in press). Accounting for the effects of levels of protection on ecological representation improves protected area network design. *Ecological Applications*.

Oldford, G., Jarníková, T., Christensen, V., & Dunphy, M. (2025). HOTSSea v1: a NEMO-based physical Hindcast of the Salish Sea (1980–2018) supporting ecosystem model development. *Geoscientific Model Development*, 18(2), 211–237. https://doi.org/10.5194/gmd-18-211-2025

Oldford, G., Cote, D., Kehler, D. G., Riefesel, G. R., & Wiersma, Y. F. (2023). FIPEX v10.4: An ArcGIS Desktop Add-in for assessing impacts of fish passage barriers and longitudinal connectivity of rivers. *SoftwareX*, 23, 101469. https://doi.org/10.1016/j.softx.2023.101469

Taft, S., **Oldford, G.**, Lilley, P. L., Oetterich, S. B., Morin, J., George, M., George, M., & Christensen, V. (2022). Reconstructing the pre-European contact shoreline of Burrard Inlet (Vancouver, BC, Canada) using participatory mapping and a two-eyed seeing approach. *Institute for the Oceans and Fisheries Research Reports, University of British Columbia*. link

Full list: goldford.github.io/publications

References available on request · Updated Nov 2026